

Humberto Sandmann



✉ hsandmann@ieee.org

🔗 hsandmann.github.io

🌀 [hsandmann](#)

📘 [hsandmann](#)

🆔 0000-0001-7758-695X

Computer scientist and educator with a D.Sc. from the University of São Paulo and a research stay at the Max Planck Institute for Dynamics and Self-Organization. My research bridges bio-inspired neural computing, nonlinear dynamics, and pattern recognition, investigating how biological neuronal mechanisms can inform computational intelligence. With more than a decade of university teaching alongside senior industry practice in software architecture, machine learning, and high-performance computing, I bring research depth and applied engineering experience into the classroom.

Research Interests

DeFi, Smart contracts and decentralized applications

Small-world networks and complex systems

Computation in large-scale systems

Artificial neural networks and deep learning

Nonlinear systems and network dynamics

Synchronization of coupled oscillators

Education

mar.2012 ↑	D.Sc. thesis	Polytechnic School of the University of Sao Paulo Spike patterns and computation in dynamical neural networks
mar.2007	doi advisor presentation supported by	10.11606/T.3.2012.tde-05092012-165022 Prof. Dr. Emilio Del Moral Hernandez slides CNPq - grant 140632/2007-7 (mar.2007 → dec.2009) $\wedge$ (sep.2010 → feb.2011)
sep.2010 ↑	D.Sc. Internship advisor	Max Planck Gesellschaft für Dynamik und Selbstorganisation Prof. Dr. Marc Timme
aug.2009	supported by	CAPES - grant BEX 4076-09-5 jan.2010 → aug.2010 Pró-Reitoria da Pós-Graduação da USP aug.2009 $\wedge$ dec.2009 (flight tickets GRU $\nrightarrow$ FRA) José Humberto Sandmann - my father aug.2009 → dec.2009 (German language classes/exams) Max Planck Institute - Networks Dynamics Group aug.2009 → nov.2009 (accommodation in Goettingen)
jun.2006 ↑	M.Sc. dissertation	Polytechnic School of the University of Sao Paulo Prediction of time series using architecture based on neuro-fuzzy systems
feb.2003	doi advisor	10.11606/D.3.2006.tde-01042009-095125 Prof. Dr. Marco Túlio Carvalho de Andrade
dec.2002 ↑	B.Sc.	Faculdade de Engenharia Industrial Computer Science Bachelor
feb.1999		
dec.1998 ↑	Escola Técnica Jorge Street Technician in Industrial Computation	
feb.1996		



Academic Appointments & Teaching

now	Insper			
↑	Sao Paulo, Brazil			
jun.2018				
	now	Associate Professor		
	↑			
	feb.2020	Platform, Microservices and APIs	Spring Cloud, Spring Boot, PostgreSQL, Redis, RabbitMQ, Kafka, Docker, K8s	2026.1, 2025.2, 2025.1, 2024.1
		Coding Interviews	Course to prepare for coding interviews.	2026.1
		Machine Learning	PCA, t-SNE, UMAP, KNN, K-Means, Decision Trees, Random Forests, Support Vector Machines, Classification, Regression, Clustering, EDA, Pre-processing, Feature Engineering	2026.1
		Neural Networks and Deep Learning	Artificial Neural Networks, Perceptron, Multi Layer Perceptron, Convolutional Neural Networks, GAN, VAE, CLIP, Stable Diffusion	2025.2
		TecWeb	JS, React	2025.2
		Cloud Computing	MAAS, Juju, OpenStack, K8s, IaaS, PaaS, Computer Networks	2025.1, 2024.2
		Design of Software	Python	2025.0, 2024.1, 2023.2, 2022.1, 2021.2, 2021.1
		Information System for Administration and Economy	Excel, VBA, Python, NumPy, Pandas, Stochastic Processes, Rol	2023.2, 2023.1, 2022.2
		Spreadsheet Automation in Excel	Excel, VBA	2021.1, 2020.1
	jan.2020	Assistant Professor		
	↑			
	aug.2018	Design of Software	Python	2019.2, 2019.1
		Foundations for Physical Communication	Python, Fourier Transform, Wavelet Transform, Signal Processing	2018.2

dec.2025	ESPM		
↑	Sao Paulo, Brazil		
jun.2014	Associate Professor		
	Machine Learning	Machine Learning, PCA, KNN, K-Means, Decision Trees, Random Forests, Support Vector Machines, Naïve Bayes, Page Rank, PySpark, Classification, Regression, Clustering, Pre-processing, Feature Engineering	2025.2
	Applied Mathematics	Propositional Logic, Boolean Algebra, Set Theory, Polynomials, Functions, Sequences, Series, Logarithm, Combinatorial Analysis, Limits, Derivatives, Linear Algebra, Matrix Algebra, Graph Theory	2025.1, 2024.1, 2023.2, 2023.1, 2022.2, 2022.1, 2021.2, 2021.1, 2020.2, 2020.1
	Oriented Object Programming	Java, Design Patterns, Docker, Spring Cloud, Microservices	2025.1, 2024.1, 2023.1, 2022.1, 2021.1
	Statistics	Statistics, Probability, Random Variables, Distributions, Sampling, Estimation, Hypothesis Testing, Regression Analysis	2023.2
	Software Quality	Software Quality, Software Testing, Software Architecture, Software Engineering, Software Development Life Cycle, Agile Methodologies, Test-Driven Development, Behavior-Driven Development, Test Automation, Continuous Integration, Continuous Delivery	2024.2, 2022.2
	Architecture and Operating Systems	Computer Architecture, Operating Systems, Binary and Hexadecimal Number Systems, Number Representation, Data Representation, Assembly Language, Memory Management, Process Management, File Systems	ESPM 16-bits 2021.1, 2020.2, 2020.1, 2019.2, 2019.1, 2018.2, 2018.1, 2017.1, 2016.2, 2016.1, 2015.2, 2015.1, 2014.2
	Artificial Intelligence and Society	Artificial Intelligence, Machine Learning, Deep Learning, Neural Networks, Computer Vision, Natural Language Processing, PCA, KNN, K-Means, GANs, Transformers, LSTMs, BERT, Ethics, Bias	2022.1 (IR), 2021.1, 2020.2, 2020.1, 2019.2
	Introduction for Artificial Intelligence	Machine Learning, Deep Learning, Neural Networks, Computer Vision, Natural Language Processing, PCA, KNN, K-Means, GANs	2020.2, 2020.1, 2019.2, 2019.1
apr.2015	Toledo Prudente Centro Universitário		
↑	Presidente Prudente, Brazil		
mar.2015	Visiting Professor		
	MBA: Big Data and Machine Learning	Big Data, Hadoop, Hive, HBase, MongoDB, Neo4j, Solr, R, Octave, Self-Organizing Maps, Bayesian Decision Trees, Multilayer Perceptron	2015.1

dec.2015	FIAP			
↑	Sao Paulo, Brazil			
mar.2012	Associate Professor			
	Distributed systems	Sockets, RMI, SOAP, Microservices, J2EE, Design Patterns		2015
	MBA: Big Data and No-SQL Modelling	Big Data, Hadoop, Hive, HBase, MongoDB, Neo4j, Solr		2014.2
	Data Structure and Algorithm II	C, Data Structure, Algorithm, Complexity, Linked List, Stack, Queue, Tree, Red-Black Tree, AVL Tree, B-Tree, Hash Table, Graph, Dijkstra Algorithm, Prim Algorithm		2013
	Algorithms	C, Programming, Algorithm, Complexity		2013, 2012.2
	Database I and II	SQL, PL/SQL, Oracle, Transaction, Concurrency Control, Locking, Deadlock, Recovery, Normalization, ER Model, Relational Model, Object-Relational Model, Trigger, View, Index, Stored Procedure, Function, Package, Cursor, Sequence, Synonym		2013, 2012
	Programming III	Pointers, C, C++, Object-Oriented Programming, Inheritance, Polymorphism, Encapsulation, Abstract Class, Interface, Exception Handling		2012
jul.2013	Universidade Cruzeiro do Sul			
↑	Sao Paulo, Brazil			
mar.2012	Associate Professor			
	Mobile Programming	Android		2012.2
	Computer Architecture	Computer Architecture, Operating Systems, Binary and Hexadecimal Number Systems, Number Representation, Data Representation, Assembly Language, Memory Management, Process Management, File Systems		2012.1
	Software Engineering	Software Engineering, Software Development Life Cycle, Agile Methodologies, Test-Driven Development, Behavior-Driven Development, Test Automation		2012.1
	Introduction to Programming	C, Programming, Algorithm		2012.1
	Database II	SQL, PL/SQL, Oracle, Transaction, Concurrency Control, Locking, Deadlock, Recovery, Normalization, ER Model, Relational Model, Trigger, View, Index, Stored Procedure, Function, Package, Cursor, Sequence		2013.1, 2012.1

jul.2011	Universidade Nove de Julho		
†	Sao Paulo, Brazil		
mar.2011	Temporary Professor		
	Computer Architecture	Computer Architecture, Operating Systems, Binary and Hexadecimal Number Systems, Number Representation, Data Representation, Assembly Language, Memory Management, Process Management, File Systems	2011.1
	Introduction to Programming	C, Programming, Algorithm	2011.1
	Web Programming	HTML, CSS, JavaScript	2011.1
jul.2011	Polytechnic School of the University of Sao Paulo		
†	Sao Paulo, Brazil		
mar.2008	Teaching Assistant		
	Experimental Pattern Recognition, Modeling and Neurocomputing	Logic Gates, Linear Regression, Pattern Classification, UC Irvine Machine Learning Repository , NIST Data Gateway , Multilayer Perceptron, EEG Signals, PCA	2011.1*
	Modeling for Signal Processing: Neural Networks and Learning	Artificial Neural Networks, Multilayer Perceptron, Backpropagation, Forecasting, Time Series, Classification	2009.1, 2008.1*
	Laboratory of the Foundations of Computing Engineering	Computer Architecture, Assembly Language, Memory Management, Process Management, Register, Stack, Turing Machine, Finite State Machines, Von Neumann Architecture, Java, Orientated Object Programming	2008.1†

* grants from CAPES (PAE)

† grants from FAPESP

Professional Experience

mar.2025	H-Sandmann	Machine Learning, Software Architecture.
↑	Sao Paulo Area, Brazil	
feb.2006	Founder	
oct.2023	SelsanTech	Machine Learning, Software Architecture.
↑	Sao Paulo Area, Brazil	
mar.2012	Co-Founder	
jun.2019	RockSpoon	POS, ERP, Payroll, JavaEE, Postgres, Software
↑	Sao Paulo Area, Brazil - Palo Alto, United States	Architecture, OpenCV, Machine Learning,
may.2018	Senior Software Architect and Machine Learning Engineer	Computer Vision, GoLang, Android, Kotlin.
jul.2016	Banco Votorantim	RoE, Banking Business, High-Performance
↑	Sao Paulo Area, Brazil	Computing, J2EE, Spring Boot, Automating
oct.2015	Contractor: Senior Consultant for Computational Optimization	Testing, Jenkins.
nov.2011	TecBan	ATM, J2EE, InMemory Database, High-
↑	Sao Paulo Area, Brazil	Performance Computing.
aug.2011	Contractor: Senior Consultant for Computational Optimization	
jun.2007	ABN Amro Bank	Stock Market, Trading, Core Banking, Software
↑	Sao Paulo, Brazil	Architecture, Cryptography, HSM, BouncyCastle,
feb.2006	Contractor: Senior Software Engineer/Architect	PKI, PCKS, Java, Servlet, Database.
jan.2006	Politec	J2EE, Software Quality, Software Architecture,
↑	Sao Paulo, Brazil	Banking Business.
aug.2005	Senior System Analyst	
jul.2005	EverSystems	PKI, BouncyCastle, Cryptography, HSM, Core
↑	Sao Paulo, Brazil	Banking, Stock Market, Java, Servlet, Database,
jul.2004	Senior Software Engineer/Architect - BankBoston - CitiBank - Elly Lilly	Communication Protocol, Synchronization and
		Replication.
mar.2003	Central Online	Object-Oriented Programming, LMS .
↑	Sao Paulo, Brazil	
sep.2002	Senior System Analyst	
sep.2002	Datamace	Payroll, ERP, Software Architecture.
↑	Sao Paulo Area, Brazil	
oct.1999	Senior System Analyst	
oct.1999	Heraeus Electro-Nite	Compliance, ERP, PCP.
↑	Sao Paulo Area, Brazil	
mar.1998	Computer Programmer	

Publications

Selected peer-reviewed publications. Full and up-to-date list: [ORCID 0000-0001-7758-695X](#).

E. Del-Moral-Hernandez; H. Sandmann; G. Araujo; **Context dependent pattern recognition - A framework for hybrid architectures bridging chaotic neural networks based on Recursive Processing Elements and symbolic information**. IEEE Transactions on Neural Networks, 2009. doi: [10.1109/IJCNN.2009.5179061](#).

L.A. da Silva; H. Sandmann; E. Del-Moral-Hernandez; **A self-organizing architecture of recursive elements for continuous learning**. IEEE Transactions on Neural Networks, 2008. doi: [10.1109/IJCNN.2008.4634190](#).

E. Del-Moral-Hernandez; H. Sandmann; L.A. da Silva; **Pattern recovery in networks of recursive processing elements with continuous learning**. IEEE Transactions on Neural Networks, 2004. doi: [10.1109/IJCNN.2004.1379877](#).

Grants & Funding

CNPq	Doctoral fellowship — grant 140632/2007-7	mar.2007 → dec.2009 ∧ sep.2010 → feb.2011
CAPES	Doctoral internship abroad (Max Planck Institute) — grant BEX 4076-09-5	jan.2010 → aug.2010
CAPES (PAE)	Teaching assistantship fellowship — Polytechnic School, USP	2008, 2011
FAPESP	Teaching assistantship fellowship — Polytechnic School, USP	2008

Academic Service & Memberships

IEEE – Institute of Electrical and Electronics Engineers

Member #60216475

- **Computational Intelligence Society (CIS)** – member since 2004; Brazilian committee member (apr.2009 → aug.2009
Λ oct.2010 → mar.2012)
- **Signal Processing Society (SPS)** – 2005 → 2009
- **Engineering in Medicine and Biology Society (EMBS)** – 2005 → 2009

INNS – International Neural Network Society

Student member #3951

Languages

Portuguese	native
English	advanced
German	B1